The All-Stage $k = 3$ State-Return Induction
A Proof Summary

Abstract

We summarize the completed induction step for the $k = 3$ phase-type moment inequality

$$(n+2)\,\mathbb{E}Z^2\,\mathbb{E}Z^4 \geq (n+3)(\mathbb{E}Z^3)^2.$$

The proof uses the state-return decomposition, monotonicity in the fourth moments, a positive-definite quadratic minimization in the two third moments, and a scale-free atom/Erlang parametrization of the sharp second-moment boundary. The resulting corner and two face conditions are proved by exact Bernstein and discriminant certificates. The complete algebraic induction step for every $n \geq 4$ is machine-checked in Lean.

1 Statement and scope

For a nonnegative random variable Z write $m_j(Z) = \mathbb{E}Z^j$. The $k = 3$ inequality at phase order n is

$$(n+2)m_2(Z)m_4(Z) - (n+3)m_3(Z)^2 \geq 0. \tag{1}$$

The induction hypothesis applies to the two phase-type excursions of order at most $n-1$. If their first four moments are

$$(u, A, C, E) = (m_1(U), m_2(U), m_3(U), m_4(U)), \qquad (v, B, D, F) = (m_1(V), m_2(V), m_3(V), m_4(V)),$$

then the required lower bounds are

$$nu^2 \leq (n-1)A, \qquad nv^2 \leq (n-1)B, \tag{2}$$

$$(n+1)A^2 \leq nuC, \qquad (n+1)B^2 \leq nvD, \tag{3}$$

$$(n+2)C^2 \leq (n+1)AE, \qquad (n+2)D^2 \leq (n+1)BF. \tag{4}$$

The all-stage Lean theorem proves the scalar renewal implication in the genuine cyclic region. Combined with the standard state-return decomposition and its degenerate boundary cases, this is the induction step for (1). The small orders are handled separately by the direct two-state and canonical three-state calculations.

2 Renewal reduction

After splitting at a chosen transient state, the absorption time has the form

$$Z = (E_0 + V_0) + \sum_{j=1}^{N}(E_j + U_j),$$

where the E_j are exponential holding times, U_j are return excursions, V_0 is the terminal excursion, and N is geometric. Let $x \geq 0$ be the mean holding time and let $r > 0$ be the mean number of returns. Define the moments of the return and terminal kernels by

$$a_1 = x + u, \qquad a_2 = 2x^2 + 2xu + A,$$
$$a_3 = 6x^3 + 6x^2u + 3xA + C, \qquad a_4 = 24x^4 + 24x^3u + 12x^2A + 4xC + E,$$

and similarly b_j with (u, A, C, E) replaced by (v, B, D, F). The renewal moments $M_j = m_j(Z)$ obey

$$\begin{aligned} M_1 &= b_1 + ra_1, \\ M_2 &= b_2 + ra_2 + 2ra_1M_1, \\ M_3 &= b_3 + ra_3 + 3ra_2M_1 + 3ra_1M_2, \\ M_4 &= b_4 + ra_4 + 4ra_3M_1 + 6ra_2M_2 + 4ra_1M_3. \end{aligned} \tag{5}$$

Thus the desired gap is

$$\mathcal{G}_n = (n+2)M_2M_4 - (n+3)M_3^2.$$

The coefficient of each of E and F in $\mathcal{G}_n$ is nonnegative. Hence (4) permits the sharp substitutions

$$E_0 = \frac{n+2}{n+1}\frac{C^2}{A}, \qquad F_0 = \frac{n+2}{n+1}\frac{D^2}{B}. \tag{6}$$

It is therefore enough to prove nonnegativity of the reduced gap

$$G(C, D) = \mathcal{G}_n\big|_{E=E_0,\, F=F_0}.$$

3 The convex third-moment layer

For fixed n, r, x, u, A, v, B, the function G is quadratic in (C, D). Put

$$H = \begin{pmatrix} H_{AA} & H_{AB} \\ H_{AB} & H_{BB} \end{pmatrix} = \nabla^2_{C,D}G, \qquad K = r^2u^2 + 2r^2ux + r^2x^2 + ruv + 2rux + rvx + 2rx^2 + vx + x^2.$$

The entries used in the formal proof are

$$\begin{aligned} H_{AA} &= \frac{2r\{Ar + (n+2)^2B + 2(n+2)^2K\}}{(n+1)A}, \\ H_{AB} &= -2(n+3)r, \\ H_{BB} &= \frac{2\{(n+2)^2Ar + B + 2(n+2)^2K\}}{(n+1)B}. \end{aligned}$$

For $r, u, v, A, B > 0$ and $x \geq 0$, both diagonal entries are positive, while

$$\det H = \frac{4(n+2)^2r}{(n+1)^2AB}(Ar + B + 2K)\big(Ar + B + 2(n+2)^2K\big) > 0. \tag{7}$$

Thus G is strictly convex on the feasible quadrant for (C, D).

Set the sharp lower endpoints from (3) to be

$$C_0 = \frac{n+1}{n}\frac{A^2}{u}, \qquad D_0 = \frac{n+1}{n}\frac{B^2}{v},$$

and write $C = C_0 + s$, $D = D_0 + t$, where $s, t \geq 0$. If

$$g = G(C_0, D_0), \qquad g_A = \partial_C G(C_0, D_0), \qquad g_B = \partial_D G(C_0, D_0),$$

then the exact Taylor formula is

$$G(C_0 + s, D_0 + t) = g + g_A s + g_B t + \frac{1}{2}\big(H_{AA}s^2 + 2H_{AB}st + H_{BB}t^2\big). \tag{8}$$

The minimum of (8) over $s, t \geq 0$ has four possible KKT locations: the corner, one of the two open faces, or the interior. Define

$$\begin{aligned} S_A &= H_{AA}g_B - H_{AB}g_A, & F_A &= 2H_{AA}g - g_A^2, \\ S_B &= H_{BB}g_A - H_{AB}g_B, & F_B &= 2H_{BB}g - g_B^2. \end{aligned}$$

It is enough to establish

1. $g \geq 0$ at the corner;
2. if $g_A < 0$ and $S_A \geq 0$, then $F_A \geq 0$;
3. if $g_B < 0$ and $S_B \geq 0$, then $F_B \geq 0$;
4. the alternative $S_A < 0$ and $S_B < 0$ cannot occur.

This replaces the failed attempt to prove convexity in the second moments: only the third-moment layer is convex, and its boundary minima must be treated explicitly.

4 Sharp second-moment coordinates

The inequalities (2) have a convenient sharp-mixture parametrization. Let

$$p = \frac{nu^2}{(n-1)A}, \qquad q = \frac{nv^2}{(n-1)B}; \qquad 0 < p, q \leq 1.$$

These are the weights of an atom/Erlang-$(n-1)$ representation. After one positive global rescaling, introduce

$$r = \frac{a}{pY}, \qquad x = \frac{pY(1-a)}{pY + a}, \qquad u = pY, \qquad A = \frac{npY^2}{n-1}, \qquad v = qT, \qquad B = \frac{nqT^2}{n-1}, \tag{9}$$

with

$$0 < a \leq 1, \qquad T, Y > 0, \qquad 0 < p, q \leq 1.$$

Conversely, every nondegenerate point satisfying (2) can be put in this form. The normalized sharp gap and KKT quantities are independent of p, except for harmless positive powers. We denote them by

$$\widehat{g}, \quad \widehat{g}_A, \quad \widehat{g}_B, \quad \widehat{S}_A, \quad \widehat{S}_B, \quad \widehat{F}_A, \quad \widehat{F}_B.$$

For example, the exact scaling identities include

$$g = \widehat{g}, \qquad g_A = \frac{2a}{Yp(n-1)^2}\,\widehat{g}_A, \qquad g_B = -\frac{2}{(n-1)^2}\,\widehat{g}_B, \qquad p^2 F_A = \widehat{F}_A, \qquad F_B = \widehat{F}_B.$$

All omitted factors in these identities are strictly positive on (9), so the relevant signs are preserved (with the shown sign reversal for g_B).

5 The sharp corner and the interior alternative

After clearing the positive denominator $(n-1)^4$, the corner gap becomes a polynomial $P_n(a,T,Y,q)$. It is quadratic in q. Its Bernstein expansion on $q \in [0,1]$ reduces positivity to the endpoints and a middle coefficient. The $q=0$ endpoint is expanded in Bernstein form in a. The $q=1$ endpoint is again expanded in a, with the terminal scale split at $T = n-1$:

$$T \le n-1 \quad \text{and} \quad T \ge n-1.$$

Writing $m = n-4 \ge 0$ and, respectively,

$$T = (n-1)t \quad (0 \le t \le 1), \qquad \text{or} \qquad T = n-1+z \quad (z \ge 0),$$

turns each coefficient into a sum of nonnegative monomials or a nonnegative quadratic certified by its discriminant. Consequently

$$\widehat{g}(n,a,T,Y,q) \ge 0 \quad (n \ge 4, 0 \le a \le 1, T, Y \ge 0, 0 \le q \le 1). \tag{10}$$

The two strict interior side conditions have opposite normalized signs. The return-side condition becomes positivity of an affine expression $R(n,a,T,Y,q)$, whereas the terminal-side condition is represented by a quadratic $S(n,a,T,Y,q)$. Exact elimination gives

$$\alpha(n,a,T,q)^2 S = \beta(n,a)R^2 + Q_1(n,a,T,q)R + Q_0(n,a,T,q), \tag{11}$$

where $\alpha > 0$ in the active region and $\beta > 0$. Separate exact certificates for the cases determined by the sign of the affine intercept show

$$R > 0 \quad \Longrightarrow \quad S \ge 0.$$

But a strict interior minimizer would require both normalized side expressions to be negative. The implication above makes that impossible. Hence the fourth KKT alternative is eliminated for every $n \ge 4$.

6 The return face

The return-face condition is needed only when the corresponding gradient is active:

$$\widehat{g}_A < 0, \qquad R \le 0.$$

Clearing positive denominators gives the exact identity

$$Y^3 n(n+1)(n-1)^4 \widehat{F}_A = 4a\, N_A, \qquad N_A = (n+2) M_2^{\text{num}}\, P_A(n, aY, T, q), \tag{12}$$

where $M_2^{\text{num}} \ge 0$ and P_A is cubic in q.

There are three reductions.

1. If $T \le n-1$, a Bernstein expansion in q proves $P_A \ge 0$ unconditionally.
2. Suppose $T \ge n-1$ and the current affine slope M of the active return interval is nonpositive. After setting $m = n-4$ and $z = T-(n-1)$, a root/surplus parametrization maps the domain to $m, s, X \ge 0$ and $v \in [0,1]$. The X-axis is covered by

   $$[0,1], \quad [1, m^2+39], \quad [m^2+39,\, 2m^2+6m+60], \quad [2m^2+6m+60, \infty).$$

   On each interval, the transformed polynomial has an exact Bernstein-positive or monotonicity certificate.

3. If $M > 0$, the side condition $R \le 0$ produces a projective coordinate $w \in [0,1]$. The active-gradient condition remains in the precise form needed at that point. A low/high chart split, followed in the high chart by two projective ratio charts, proves

$$\widehat{g}_A < 0 \quad \Longrightarrow \quad P_A \ge 0.$$

The individual chart polynomials are certified by Bernstein coefficients and, where a middle coefficient changes sign, by a quadratic discriminant.

Together these cases prove

$$\widehat{g}_A < 0, R \le 0 \quad \Longrightarrow \quad \widehat{F}_A \ge 0. \tag{13}$$

7 The terminal face

The terminal-face numerator admits a particularly useful factorization:

$$N_B = (n+2)\, L(n,a,T,Y,q)\, P_B(n,a,T,Y,q), \tag{14}$$

where $L > 0$ and P_B is quadratic in q. Write its quadratic Bernstein form as

$$P_B = B_0(1-q)^2 + 2B_1 q(1-q) + B_2 q^2. \tag{15}$$

The coefficient B_0 is nonnegative by a cubic Bernstein expansion in a. The other two coefficients are controlled by the endpoint values of the affine terminal gradient. The initial range $T \le 1$, and then the larger compact strip $T \le 9$, are handled by direct Bernstein and positive-semidefinite certificates. The right endpoint B_2 is unconditional whenever its gradient condition is active.

Only the case $T \ge 9$ for the middle coefficient B_1 remains. Set

$$m = n-4, \qquad t = T-1 \ge 8, \qquad X = aY.$$

The active left-gradient condition is affine in t and yields the time budget

$$aC(m,X)(t-8) \le h(m,a,X), \qquad C(m,X) > 0, \quad h(m,a,X) \ge 0. \tag{16}$$

Let $U = h/(aC)$ and $w = (t-8)/U$. Then $0 \le w \le 1$, and the middle coefficient is a quadratic Bernstein polynomial

$$B_1(t) = C_0(1-w)^2 + 2C_1 w(1-w) + C_2 w^2. \tag{17}$$

Exact certificates give $C_0, C_2 \ge 0$. If $C_1 \ge 0$, positivity is immediate. If $C_1 < 0$, the required condition is

$$C_0 C_2 - C_1^2 \ge 0. \tag{18}$$

After multiplying by a positive square, the left side of (18) equals

$$h(m,a,X)^2 (m+6) J(m,a,X).$$

The final exact identity is

$$J = (-N)Q + \sum_{i=0}^{4} \binom{4}{i} R_i(m,X)\, a^i (1-a)^{4-i}, \tag{19}$$

where N is a positive multiple of C_1, $Q \ge 0$, and every $R_i(m,X) \ge 0$ for $m, X \ge 0$. Thus $C_1 < 0$ implies $N \le 0$, so every term on the right of (19) is nonnegative. This proves (18), hence $B_1 \ge 0$, and finally

$$\widehat{g}_B > 0, S \ge 0 \quad \Longrightarrow \quad \widehat{F}_B \ge 0. \tag{20}$$

8 Assembly of the induction step

Equations (10), (13), and (20), together with the impossibility of the strict interior KKT alternative, apply the convex-quadrant lemma to (8). Therefore $G(C, D) \geq 0$ for all C, D satisfying (3). Monotonicity in the two fourth moments then restores the actual E, F from their lower bounds (6), proving (1) for the renewal law.

Equivalently, the final normalized statement is the following.

Theorem 1 (all-stage $k = 3$ renewal induction)**.** *Let $n \geq 4$, $0 < a \leq 1$, $T, Y > 0$, and $0 < p, q \leq 1$. Put*

$$r = \frac{a}{pY}, \quad x = \frac{pY(1-a)}{pY+a}, \quad u = pY, \quad A = \frac{npY^2}{n-1}, \quad v = qT, \quad B = \frac{nqT^2}{n-1}.$$

If

$$(n+1)A^2 \leq nuC, \qquad (n+1)B^2 \leq nvD,$$

then the fourth-moment-reduced renewal gap satisfies

$$G(C, D) \geq 0.$$

Consequently, if the two excursion fourth moments also satisfy (4)*, then the full renewal moments satisfy*

$$(n+2)M_2M_4 \geq (n+3)M_3^2.$$

9 Formal verification map

The principal Lean files are:

- `K3AllStages.lean`: renewal moments, fourth-moment monotonicity, the positive Hessian, the quadrant KKT reduction, mixture coordinates, the sharp corner, and exclusion of the interior point;
- `K3AllStagesTail.lean`, `K3AllStagesMiddleLow.lean`, and `K3AllStagesMiddleHigh.lean`: the nonpositive-slope return-face charts;
- `K3AllStagesPositiveSlope*.lean`: the positive-slope return-face charts;
- `K3AllStagesTerminalFace.lean` and `K3AllStagesTerminalFaceRemainingCert.lean`: the terminal face and the large-time determinant certificate;
- `K3AllStagesComplete.lean`: the final assembly, whose main theorem is `excursionReduced Gap_mixture_coordinates_nonneg`.

Every polynomial identity is checked over the rationals by Lean's exact ring normalizer; every positivity step is reduced to explicit nonnegative factors, Bernstein coefficients, or a 2×2 determinant. No numerical approximation enters the proof.